

add_sediment_fraction(name, diameter)
for each grain size

1 What is the bed made of?

Sand, gravel, or mud that moves as flocs
set_bed_material('noncohesive')
entrainment='smith_mclean' (default),
or 'de_leeuw' for sand rivers and deltas

Clay or silt that binds
set_bed_material('cohesive', tau_crit=...)
tau_crit in Pa, ideally from a jet test

2 How does it move?
Rouse number $Z = w_s / (0.41 u^*)$

Z above 2.5: bedload only
set_bedload('wong_parker_eq24')

Z from 0.8 to 2.5: both
set_bedload(...) as well

Z below 0.8: suspension
no bedload (the default)

3 Is the suspension stratified?

Yes, Z above about 0.1
set_deposition(near_bed='rouse')

No, well mixed
keep $d^* = 1$ (the default)

4 What sets the bed shear?

Open channel or floodplain
the defaults: 'quadratic_drag'
with the domain's Manning n

Vegetation
set_vegetation_drag(...,
sediment_shear='bed')

5 How much of the bed can erode?

A layer over rock, or a lining
set_erodible_base(depth=...)

Only part may scour
set_erodible_region(...)

Banks or holes that should slump
set_angle_of_repose(35.0)

6 Several grain sizes, and
the bed sorts them?

Yes: mud over sand, armouring
set_bed_composition(fractions,
active_layer=0.01)
after set_erodible_base

7 Is the bed much slower
than the flow?

Days to years of bed change
set_sediment_parameters(
morphological_factor=M)

A fixed bed (analytic check)
bed_evolution=False

print(domain.sediment_summary())
then calibrate against data